

3(a)	$s = \frac{1}{2}at^2$	C1
	$57 = \frac{1}{2} \times 9.81 \times t^2$ and $t = 3.4$ (s)	A1
3(b)	horizontal distance = 41×3.4 = 140 m	A1
3(c)	(displacement) ² = $57^2 + 140^2$	C1
	displacement = $(57^2 + 140^2)^{0.5}$ = 150 m	A1
3(d)	straight line from the origin with positive gradient	B1
3(e)	$(1480 - m) \times 0.340 = m \times 41.0$	C1
	$m = 12.2$ kg	A1
	or	
	$m_c 0.34 = m_b 41$ and $m_c + m_b = 1480$	(C1)
	$m_c = (41 / 0.34)m_b$ $(41 / 0.34)m_b + m_b = 1480$ $m_b = 12.2$ kg	(A1)
3(f)	acceleration (of free fall) is unchanged/is not dependent on mass	M1
	(so) no change (to the graph)	A1